SDHMA

Std - X **Preliminary Examination** 2025 -26 **Marks - 40**

Time - 2.00 Hrs. **Sub - Mathematics - I** **Pages - 4**

Note - 1) All questions are compulsory.

2) Use of calculator is not allowed.

3) In case of MCQ's Q. No. 1(A) only the first attempt will be evaluated and will be given credit.

4) For every MCQ, Four alternative (a), (b), (c) or (d) of answers are given altenalternative of correst answer is to be written in frount of the subquestion number.

5) Draw figure if necessary with answers.

1. A) Choose the correct alternative to complete the sentences. **4**

1) If $99x + 101y = 499$; $101x + 99y = 501$ then $x + y$ = ...

a) 1000 b) 5

c) 1 d) 200

2) If the equation $3y^2 = 15y$ is solved by factor method, the solution will be .....

a) 5 b) 0

c) 0 or 5 d) None of these

3) The sum of all terms in the A.P 2, 4, 6.....20 is .....

a) 110 b) 40

c) 20 d) 100

4) A die is thrown. The probability of getging number on upper face more than 3 is .....

a) 3 b) 6

c) $\frac{1}{2}$ d) 2

B) Solve the following questions. **4**

1) State how many alphanumericals are there in a GSTIN. Give an example.

2) Prepare a sample space for selecting a day of a week at a time randomly.

3) The length of rectangle is x and its breath is x. Form a linear equation if the perimeter of rectangle is 50 cm.

4) Find t_n, if a = 3.5 and d = 0

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2. A) Complete the following activities. (Any 2) **4**

1) The price of an article is ₹ 500. If the rate of GST is 12% then to find the purchase price for the customer. Complete the following activity.

Price of article = ₹ 500

Rate of GST = 12%

Amount of CGST = Amount of SGST = $\frac{6}{100}$ X [ ] = ₹ [ ]

Amount of SGST =

Total GST = ₹ [ ]

∴ Purchase price of article = ₹ [ ] + 60

= ₹ 560

2) Complete the table to draw the graph of x + y = 3

x	3	[ ]	[ ]
y	[ ]	5	3
(x, y)	(3, 0)	[ ]	(0, 3)

3) Sum of roots of a quadratic equation is 5 and sum of their cubes is 35. Complete the following activity.

Let α and β are the roots.

Here $\alpha + \beta = 5$ $\alpha^3 + \beta^3 = 35$

Now $\alpha^3 + \beta^3 = (\alpha + \beta)^3 -$ [ ] ..... formula

[ ] $= (5)^3 - 3\alpha\beta\,(5)$

∴ $\alpha\beta =$ [ ]

∴ The quadratic eqⁿ is in the from

$x^2 - (\alpha + \beta)\,x + \alpha\beta = 0$

∴ Required quadratic equation is [ ]

B) Solve the following. (Any 4) **8**

1) The table shows the time in hours spend by 50 students of class 10th for study and the number of students. Find the mean of the time in hours taken by the students by direct mean method.

Time (Hrs)	0 - 2	2 - 4	4 - 6
Number of students	15	15	20

2) The amount of ₹ 60,000 was invested to purchase the share of face value ₹ 100 when the market price was ₹ 120. If the dividend of 10% was declared. Find the dividend.

3) Find the value of K if the roots of the quadratic equation $3y^2 + Ky + 12 = 0$ are real and equal.

4) The difference between the two complementary angles is 10^0. Find the measure of larger angle.

5) Find the sum of first ten terms of an A.P. whose first term is 6 and comman difference is 10

3. A) Complete the following activity. (Any 1) **3**

1) There are 15 tickets in a box, bearing with the numbers from 1 to 15. on each ticket. One ticket is to be drawn randomly from the for box for finding the probability of the following events, Complete the activity.

i) The number is an even ii) The number is multiple of 3

i) S = { }

$\therefore$ n (s) = 15

ii) Event A = The number is an even

$\therefore$ A = { }

$\therefore$ n (A) = [ ]

$\therefore$ P (A) = $\frac{n(A)}{n(s)}$ = [ ]

iii) Event B - The number is multiple of 3

$\therefore$ B = { }

$\therefore$ n (B) = 5

$\therefore$ P (B) = $\frac{[\ ]}{n(s)} = \frac{1}{3}$

2) Prepare a Business to Business (B2B) tax invoice as per the details given below. Complete the following activity.

Supplier - M/S Golden Stores.

Address - Navi Peth Solapuer

State - Maharashtra

Date - 06. 01. 2025 Invoice No - 215

GSTIN - 27 PQRST 2914J1ZS

Sr. No.	HSN Code	Name of item	Rate (₹)	Quantity	Cost	CGST 9%	SGST 9%	Total amount
i)	2109	Chockolate	50	4	[ ]	[ ]	[ ]	₹ 236
ii)	2201	Ice-cream	100	3	300	[ ]	[ ]	[ ]
						Total = ₹ 590		

B) Solve the following. (Any 2) 6

1) The product of Rohit's age 3 years ago and 4 years hence is 144. Find his present age.

2) Solve by Cramer's method $5x - 6y + 30 = 0$; $5x + 4y - 20 = 0$

3) The distance travelled by 38 buses of public transport in one day are given in the table. Find the median of the distance travelled.

Distance travelled (km)	100 - 110	110 - 120	120 - 130	130 - 140
Number of buses	15	05	10	08

4) A two digit number is formed with digits 2, 3, 4, 5, 6 without repeating the digit. What is the probability of the following event that the number formed is

i) a perfect square ii) greater than 50

4. Solve. (Any 2) 8

1) A questionare was solved by 50 students of class 10th to know their emotional intelligence score. The marks obtained by them are given in the table below. Show this information by histogram and frequency polygon on the same graph. https://www.maharashtrastudy.com

Marks obtained	40 - 49	50 - 59	60 - 59	70 - 79
No of students	10	14	17	09

2) Pratik takes 8 hours to travel 36 km downstream and return to the same spot. The speed of boat in still water is 12 Km. per hour. Find the speed of water current.

3) In an auditorium, there are 40 rows of chairs. There are 100 chairs in 40th row, 98 in 39th , 96 in 38th and 50 on. Then find

i) Number of chairs in 15th row

ii) Total number of chairs in the auditorium

iii) The number of chairs in 50th row if there were 50 rows of chairs.

5. Solve. (Any 1) 3

1) The measures of the angles of triangle are in the ratio 3 : 4 : 5. Find the measure of each angle and show them using Pie diagram.

2) Solve. $\frac{x + y - 2}{5} = \frac{2x - y}{6} = \frac{2x + y}{10}$

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